DSA06 - Data Handling and Visualization

Activity 5: Normality Detection using Q-Q Plots | Slot D

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Part A (16 marks)

Chart 1: Commute time (minutes), n = 30

Normal? (Yes / No)	No
Shape (from the list)	Right-skewed (with high outliers)
Evidence in the plot	Points curve above the red line at the upper end (about 65 and 80 minutes), while the middle points follow the line.

Chart 2: Lab experiment duration (minutes), n = 50

Normal? (Yes / No)	Yes (approximately)
Shape (from the list)	Normal
Evidence in the plot	Most points lie close to the red line from end to end, with only small wiggles at the extremes.

Chart 3: Daily study time (minutes), n = 60

Normal? (Yes / No)	No
Shape (from the list)	Normal with outlier(s)
Evidence in the plot	Central points hug the red line, but two points at the top right (about 280 and 300) lie far above it.

Chart 4: Plant height (cm), n = 120

Normal? (Yes / No)	No
Shape (from the list)	Heavy-tailed
Evidence in the plot	Middle follows the line, but the lowest points fall below the line and the highest points rise above it (S-shaped ends).

Part B (4 marks)

B1. Chart 4 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods).

Ans: Chart 4 is heavy-tailed: the extreme low and high points lie away from the red line, so there are more extreme values than a normal distribution would give. Remedy: use robust methods (median and IQR / trimmed mean) instead of mean and SD.

B2. Fill in the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____

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ggplot(df, aes(sample = value)) + stat_qq() + stat_qq_line()
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